



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

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Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

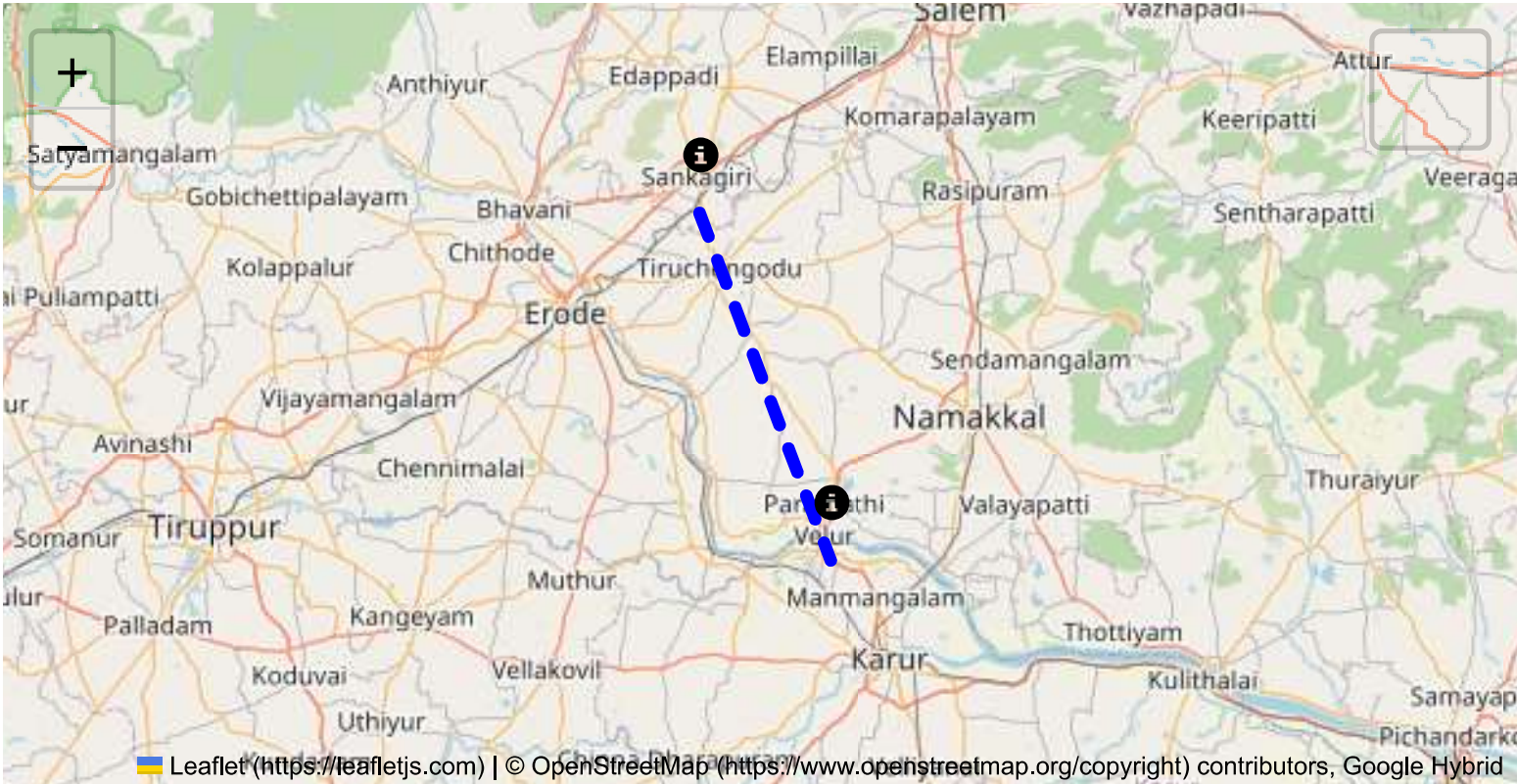
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

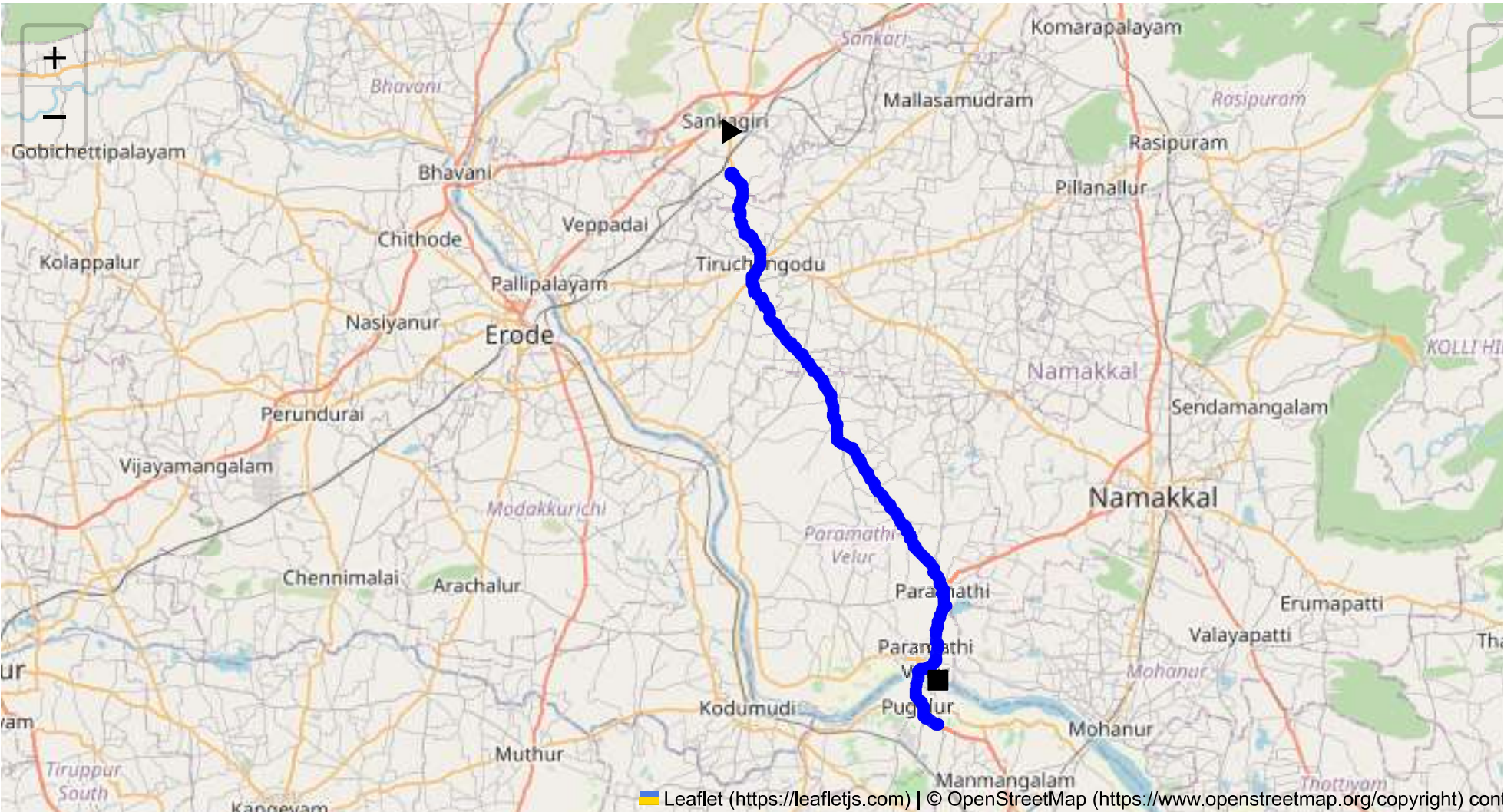
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 49.92 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours
Start: (11.4381, 77.8734)
End: (11.064344, 78.017296)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

1. Overview of the Route Map:

This route spans approximately 49.92 kilometers from Sangagiri to the specified location on NH 44 in Tamil Nadu. The journey primarily involves travelling on state highways and smaller local roads, with

significant portions likely utilizing National Highway 44 (NH 44) as it nears the destination.

2. Typical Weather Conditions and Potential Hazards:

- **Weather Patterns:** Tamil Nadu typically experiences a tropical climate with high temperatures year-round. Monsoons occur from June to September with the northeast monsoon bringing additional rainfall from October to December.
- **Weather-Related Hazards:** Heavy rains during monsoons can lead to waterlogged roads and reduced visibility. Fog is less common but can occasionally occur.

3. Traffic Patterns:

- **Peak Hours:** Expect increased traffic, particularly during early mornings (8-10 AM) and late afternoons to early evenings (5-7 PM) due to local commuting patterns.
- **Congestion-Prone Areas:** Urban centers near towns and intersections with major roads can experience significant slowdowns, particularly during peak hours.

4. Assessment of Road Quality and Infrastructure:

- **Road Quality:** The condition of the roads can vary; NH 44 is generally well-maintained, whereas smaller state and local roads may have potholes or uneven surfaces.
- **Infrastructure:** Adequate signage is generally available along NH 44, but secondary roads may lack proper road markings and signs.

5. Suggestions for Alternative Routes in Emergencies:

- **Alternative Routes:** In case of road blockages or accidents, consider using nearby parallel state highways. Local knowledge of smaller roads can also provide detours.
- **Local Roads:** Familiarity with local roads might offer quick escapes, though they might not support heavy trucks well.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- **Regulations:** The transport of hazardous materials is subject to stringent state and national regulations, including specific times for permissible travel (often restricted during peak traffic hours).
- **Permits:** Ensure all necessary permits are secured for handling and transporting such materials.

7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

- **Accidents:** Historically, this region has experienced sporadic incidents involving heavy vehicles, often due to poor road conditions and driver error.
- **Material Spills:** There have been few notable incidents involving hazardous materials, which generally occurred due to improper handling or accidents.

8. Environmental Considerations and Sensitive Areas:

- Environmental Zones:** Be aware of agricultural zones and densely populated areas where spills or accidents could have significant environmental or social impacts.

9. Communication Coverage:

- Coverage:** Generally good along NH 44 and in urban centers, but more remote areas may have spotty reception leading to potential dead zones.

10. Estimated Emergency Response Times:

- Near Urban Areas:** Approximately 20-30 minutes.
- Rural Segments:** Response times can be longer, potentially exceeding 45 minutes, particularly when further from major towns or cities.

12. Overall Summary of Risk Assessment:

- Overall Risk:** Moderate. Main risks involve weather-related travel during monsoons, road quality on minor roads, and potential communication dead zones.
- Mitigation Strategies:** Emphasize regular weather checks for the route, choose routes with better road conditions when possible, and equip vehicles with reliable communication equipment.

Conclusively, while the primary risks involve environmental and road condition factors, adhering to regulations and enhancing driver preparedness can significantly mitigate potential hazards on this route.

Recommendations for Drivers:

- Conduct routine vehicle inspections focusing on braking systems, particularly in wet conditions.
- Stay updated on weather forecasts, especially during monsoon seasons.
- Maintain strong communication with dispatch teams, especially when traversing remote areas prone to dead zones.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	Medium	11.16764, 78.01565	30 KM/Hr
8	Turn	High	11.15573, 78.02176	15 KM/Hr
9	Turn	High	11.14545, 78.02208	15 KM/Hr
10	Turn	High	11.14480, 78.02291	15 KM/Hr
11	Blind Spot	Blind Spot	11.06358, 78.01751	10 KM/Hr
12	Turn	High	11.06408, 78.01744	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.15573, 78.02176



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.06358, 78.01751



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.06408, 78.01744